ELSEVIER

To boldly go where no one has gone before: Sulfur concrete, a promising construction material fulfilling the demands for a sustainable future on celestial objects: A review

RESEARCH: Review

Q. Wang, D. Snoeck*

BATir Department, École Polytechnique de Bruxelles, Université Libre de Bruxelles (ULB), 50 F.D. Roosevelt Ave, CP 194/02, 1050 Brussels, Belgium

The need to build a long-term or even permanent base is now a significant concern with the development of the exploration of extraterrestrial celestial bodies. Sulfur concrete was first proposed as a new building material in the 20th century. Recently, sulfur concrete has attracted much interest, as sulfur is considered one of the most accessible resources on the Moon and Mars, thanks to the in-situ resource utilization methodology. In addition, sulfur concrete is one of the most promising building materials for improving terrestrial sustainability or extraterrestrial exploration. So far, reviews have only focused on developing sulfur concrete and extraterrestrial building materials. This review paper summarizes the history of sulfur concrete development and different modified sulfur concretes. Previous research on extraterrestrial building materials is also reviewed. The unique advantage of sulfur concrete as an extraterrestrial material is justified, as no water is used during mixing. Lunar and Martian soil simulants are also examined as possible aggregate types. Finally, further improvements are proposed to broaden the application of sulfur concrete and the corresponding treatments. The possibility of recyclability and circularity is discussed from a sustainable development point of view. This review article provides readers with a detailed overview of sulfur concrete and its history, why it is more promising and accessible as an (extra)terrestrial building material, the challenges of its future application, and corresponding treatments to overcome the obstacles.

Keywords: Sulfur concrete; Sustainability; Extraterrestrial construction materials; Circularity; Recyclability; ISRU

Introduction

Since 1961, when Russian/former Soviet Union astronaut Yuri Gagarin was the first human to fly into space, [1] the entire world accelerated the exploration of the endless universe. The nearest celestial object - the Moon - became the first target humans tried to reach. [2] Starting from the 1960 s, explorations never ceased for landing on the Moon by different countries, and the former Soviet Union and the U.S. achieved many goals with their lunar explorations and beyond. [3,4] Afterwards, further exploration of the Moon and even Mars became the goal: NASA has planned the Artemis program as the outpost for their national strategy to strengthen the leadership of the U.S. in space exploration and aeronautics, which emphasizes extending the presence of humans on the Moon and Mars, towards a sustainable long-term exploration. [5] The ESA published their space resource strategy, which proposed that by 2040, the goal should be achieved that local resources will sustain human presence on the Moon. [6] China, as one of the rising players in space exploration, launched Chang'e series lunar probes to collect and bring back samples from polar regions and explore the dark side of the Moon, while completing critical technological research on Mars

* Corresponding author.
E-mail address: Snoeck, D. (didier.snoeck@ulb.be)

1369-7021/© 2023 Elsevier Ltd. All rights reserved. https://doi.org/10.1016/j.mattod.2023.12.005

sampling and return. [7] It is thus clear that there is a general interest in exploring the use of building materials on the Moon and Mars using local resources. Especially since the Indian lander Chandrayaan-3 found sulfur at the Moon's south pole, something that had not previously been detected by orbiting probes.

Due to the renewed interest in establishing installations and possibly colonies on the Moon in 2028 [8]-2040 [6] and Mars from 2030 onwards [9,10], reliable construction materials are needed for building infrastructures on celestial objects. To achieve those ambitious goals, it is unavoidable to consider making lunar or Martian bases for better long-term exploration activities. Conventional building materials such as reinforced concrete and existing knowledge on structural behavior become unreliable when considering the Moon and Mars environment, such as a lower gravity of 16.7 % and 38 % of that on Earth, respectively, and different thermal cycles and daily temperature changes. [11] Meanwhile, it is uneconomical to transport materials from the Earth to the Moon, which is estimated to be around USD$130,000 per kg of material from the Earth to the Moon. [12,13] Tuklo [14] insists that "civil engineering is the art of directing great sources of power in nature for the use and convenience of man." This definition indicates that a feasible way to solve the problem of materials economically and logically for extraterrestrial constructions is to use resources exploited directly from them. So far, the original lunar soil samples for research are less than 400 kg. [15] Detailed research about its granulometric composition, density and porosity, cohesion and adhesion mechanical properties, etc., are completed. [16–18] Compared with lunar soil, the knowledge of Martian soil is still lacking, even though there are eight spacecraft in orbit and three on the surface of Mars when writing this review paper, [19] keeping their exploration of Mars as the preparation for further research.

As exploring the Moon and Mars continues, much research on extraterrestrial construction materials has led to inspiring findings. According to Agosto, [20] 95 % of the raw materials for conventional concrete could be obtained from the Moon. Meanwhile, McKay, [21] proposed that critical materials for concrete fabrication are much easier to get on Mars than on the Moon. Hu et al. [2] summarized five types of concrete promising for use in the Moon and Mars extraterrestrial environment. The conclusions are shown in Table 1.

Traditional hydraulic concrete is not an option as water is a precious resource on Mars and the Moon and may also become one on Earth. An interesting path that does not require cement and water is sulfur concrete. [13,41–44] It forms a high-strength and chemical-resistant material due to solidification when it cools below 140 °C, thus binding aggregates together. [42] Compared to the mentioned possibilities as construction materials on celestial objects, sulfur concrete combines the advantages of no water needed and high early compressive strength. [45,46] Meanwhile, it has the largest application area for both the Moon and Mars. [11,47–49] However, the durability of sulfur concrete applied in a vacuum, hot temperature, and temperature cycling conditions still needs to be improved to ensure long-term performance. [46,47,50].

The first part of this paper reviews sulfur concrete in a detailed way. The second part summarizes the extraterrestrial environmental conditions and constituents of the soil. Further, it discusses the In-Situ Resource Utilization (ISRU) strategy for applying sulfur concrete on the Moon and Mars. Finally, the third part will discuss the further development of sulfur concrete, focusing on better durability, integrity, and workability under extraterrestrial environments.

Sulfur concrete

Sulfur as a resource

On Earth, Sulfur is available in large quantities of more than 68 million tons each year as a by-product of the oil and gas industry.

TABLE 1

Comparison of construction materials possibly used on celestial objects, [2] linked to the application, strength, durability, and positive and negative characteristics. Compressive strength statistics from different papers indicate the ranges of different types of concrete with various components.

Concrete categories	Material composition	Applicable area	Compressive strength	Advantage	Disadvantages
Calcium Aluminate concrete	Calcium Aluminate Cement + Water + Aggregate	Lunar	25–45 MPa, [22,23] higher compressive strength by DMSI method (120 MPa). [24]	Applicable under low temperature and vacuum space.	A tremendous amount of water demands large energy consumption. [25]
Magnesia Silica Concrete	MgO + Silica + Water + Aggregate	Mars	20–64 MPa, [26,27] by adding quartz, the compressive strength increases to 87 MPa. [28]	Higher strength curing in air [29]; Low energy consumption [30]	Huge water demand: Transportation of superplasticizer from the Earth to Mars. [31,32]
Polymer Concrete	Polymer + Aggregate	Lunar	12.6–71.3 MPa [33,34]	High compressive strength, fast curing, waterless [35]	Transportation of polymers from the Earth to the Moon.
Geopolymer Concrete	Geopolymer + Alkali activator + Water + Aggregate	Lunar	10–80 MPa [36,37]	Low water consumption, high tolerant ability to temperature cycles, good vacuum stability, radiation shielding. [38–40]	Transportation of alkali-activator from the Earth to the Moon.

[51,52] It is found on the Moon as troilite (FeS) in amounts of around 0.3 wt% of the total mass. [41] Sulfur-rich ores are present as veins, and troilite blebs have been found in different Lunar samples. [43,53] Although the concentration in the total material is low, the sulfur can be extracted with 90 % yield by heating at 1000 °C, using solar energy as the heat source. [53–56] Furthermore, sulfur is a by-product of oxygen extraction, which is also needed on Lunar bases for human survival, in masses of 10 % of the oxygen mass. [44,57] Mars, on the other hand, is sulfur-rich (SO_3 up to 37 wt% and on average 6 wt%), and initial composites produced with analogous Martian materials look very promising in terms of their mechanical properties. [48] Other resources to be used are Lunar and Martian regolith. Aside from sulfur concrete, they can be processed to hydrate when in contact with water and may act as geopolymers. [58–60] However, water is not abundantly available on the Moon or Mars. Therefore, regolith may be used as aggregate and filler for sulfur concrete. High amounts of sulfur are used in a mixture (12–22 wt % and 78–88 wt% of aggregate contents [43] or even higher from 25 % to 70 %, with a previous optimum depending on the aggregates at 35 %[13]). Other ways involve the inclusion of liquid salts [61] or phosphoric acid [62] as liquid binders, again requiring transport.

History of the development of sulfur concrete

A detailed summary of the development of sulfur as a building material has been published by ACI committee 548. [63] Sulfur as an additive for construction could date back to prehistoric times, [64] one of the specific applications is to use sulfur as a hot-melt bonding agent and sealer for water vessels. [65] In the 17th century, sulfur was used to bond and fix metal with stone as an anchor method for ships. [66] After World War I, the enormous demand for sulfur led to the opening of the "Big Dome" sulfur deposit in North America, which increased the annual amount of production of sulfur to two times than before, it also created a heavy surplus of sulfur in stock, consequently, stimulated the interest for developing new applications for consuming those surplus of sulfur. [63,67,68] In 1921, Bacon and Davis [69] conducted different trials of sulfur construction materials, and they found that a sulfur mortar with 40 wt% of sulfur and 60 wt% of sand exhibited high resistant ability to acid and good mechanical properties. After that, the research on sulfur concrete progressed around the 1960 s-1980 s, especially in the United States and Canada. [68] The United States Department of the Interior's Bureau of Mines and The Sulfur Institute cooperate in 1971 to explore and develop new applications of sulfur, McBee et al., [67,70] summarized possible applications of sulfur construction materials, such as sulfur concretes, sulfur spray coatings and sulfur-extended asphalt. [67,70] The Canada Centre for Mineral and Energy Technology and the National Research Council of Canada published numerous works where the main findings are sulfur concrete is regarded as a promising construction material with much better cost-effectiveness. Still, the durability of sulfur concrete under freeze–thaw cycles is considered unreliable for long-term construction applications. [71,72,72,73] At that moment, researchers were focusing on the sulfur concrete fabricated by hot-mixing unmodified sulfur and aggregates, the results revealed one of the crucial shortages that sulfur concrete manufactured by unmodified sulfur exhibited poor durability after conducting thermal cycles. [72,74] To solve this problem, researchers started their trials to fabricate sulfur concrete by modifying sulfur for better performance in both mechanics and durability. McBee and Sullivan [75–79] were one of the first exploring modified sulfur concrete, and it was concluded that modified sulfur concrete exhibited better durability with additives. [75–79] From the 1980 s onwards, with the development of modern industries, a burst of hydrocarbon production led to stockpiles in sulfur, [80,81] which stimulated the acceleration of research on sulfur concrete.

Along with the research on sulfur concrete, its applications were also introduced into the construction field. Sulfur was not an active option for construction before the 1920 s, and a piece of recorded evidence showed that a sulfur-sand mortar was used to connect a sewer for conveying acid wastewater from a pulp mill only at that time. [71] Some ideas about potential applications of sulfur concrete were industrial floors, bridge decks, curbs, tanks, pipes or pope linings, and tunnel linings. Nowadays, sulfur concrete is mainly used in small quantities as sewer pipes, drains, culverts, sleepers, tiles, and key roadwork elements, [82,83] due to the limitations of the material. In Europe, Denmark, Sweden, Austria, and the UK pioneered sulfur construction research, [68] leading to commercial sulfur-related construction materials such as "STARcrete" [84] and "Shell Thiocrete" [85]. In 2021, Belgium began using sulfur concrete as sleepers. [86].

From an economic point of view, sulfur concrete is also competitive with conventional cement concrete. According to statistics currently available, in October 2023, the average cement price is USD\$0.22·kg^{-1} based on statistics from 22 countries, with an upper value of USD\$0.39·kg^{-1}. [87] For sulfur, prices are USD\$0.11·kg^{-1}, USD\$0.15·kg^{-1}, USD\$0.09/kg and USD \$0.11·kg^{-1} in the US, China, Qatar and Germany, respectively, in the third quarter of 2023. [88] Clearly, the price of the raw material gives sulfur concrete an impressive advantage. As a result, the commercial application of sulfur concrete is gradually increasing.

Specifically, in 1988, Leonard and Johnson [15,89] first proposed sulfur-based concrete as the construction material for the Moon since it is a water-less material and the water's existence on the Moon remained unclear. [44,90].

Characteristics of sulfur concrete (advantages, disadvantages, application)

Regarding chemical characteristics, sulfur has diverse allotropes, including intramolecular allotrope (chemical bonding between sulfur atoms), intermolecular allotrope (different sulfur molecule's arrangement), and cyclic rings. [42,52] All the allotropes differ in physical properties but are similar in most chemical properties. They can co-exist in equilibrium with definite proportions; the temperature and pressure history decide the specific form. According to the previous research, it is mentionable that the most important types among all those allotropes are rhombic (α) and monoclinic (β) sulfur, the former being stable under 95.5 with a melting point around 114 , while the latter is stable between 95.5 and its melting point around 119 . [42,52,91,92].

TABLE 2

Comparison between Portland cement (PC)-based concrete and sulfur-based concrete properties (Grade M15 and M60). [52]·

Property	Unit	PC-based concrete	Sulfur-based concrete
Time of strength gain	Time	28 days	3 h – 14 days [45,98]
Compressive strength	MPa	15–60 [45]	13–115 [48,99,100]
Tensile strength	MPa	3.31–4.43 [101]	5–7 [52]
Flexural strength	MPa	3.9–9.6 [102–104]	10–15 [52]
Freeze-thaw durability	%	60–82 [105]	300 [63]
Acid resistance [106]	% (mass loss exposed to 10 % HCl)	11.64 (21 days)	0.54 (180 days)
Permeability	$\times 10^{-11} m/s$	1.81–2.39 [107]	Negligible [108,109]
Shrinkage	$\times 10^{-6}$	−700 (1947 days) [110]	Negligible [42,52,71]
Thermal conductivity	W/m·K	0.6–3.3 [111,112]	0.128 [113]

The terrestrial advantages of using sulfur as a binder of concrete are a) the reduction of the carbon CO_2 footprint as no cement is used, [13,41–44] b) the low price of sulfur, it being a waste material from the petroleum industry, [93] c) the ability of production at freezing/elevated temperatures with an acceptable freeze-thawing resistance [48], d) the quick hardening, e) the excellent bond with, for example, porphyry aggregates, f) the high mechanical properties and high fatigue resistance [42], g) the resistance to acids and salts [42,48,63,94,95], h) the compatibility with ecosystems [48], and i) the protection against ionizing radiation [96,97]. These all support the potential of sulfur concrete in relation to its application on celestial objects. Furthermore, the most exciting feature of the material is its recyclability since it can be molten at temperatures above 120 °C. It is thus a circular material, which is also very important for the overall environmental impact of sulfur concrete for a sustainable future. The sulfur used as a binder in sulfur concrete started with unmodified sulfur. The mixture exhibited excellent performance, such as impermeability, low thermal and electrical conductivity, high resistance to acids under the action of water and sunlight, and higher mechanical properties compared with Portland concrete with Portland cement. Table 2 provides a summary for comparison between sulfur concrete and Portland cement concrete. [52].

Unfortunately, the material also has disadvantages. First, upon mixing, there is a need to control fumes and maintain a constant temperature. [114]) Furthermore, heating the material above 120 °C leads to a loss in integrity, which limits its application due to thermal constraints. Lunar temperatures at the poles are in the range of 60 °C to 220 °C, while at the equator, the range is 123 °C to 180 °C. [43] As sulfur melts at 120 °C, this imposes a problem. [42,43,47] Some authors report a maximum of 117 °C for lunar temperatures, which is still a problem. [62] The atmospheric conditions on Mars with a temperature of 35 °C or lower are more promising. [48] There are additional extraterrestrial difficulties as the environmental conditions change with a different atmosphere and gravity compared to Earth. The gravity on the Moon is one-sixth of that on Earth, and one Lunar day equals 29 Earth days. [43] Vacuum is another major problem. In a vacuum, sulfur may sublimate, causing the material to disappear. [46,47] In literature, a continuous weight loss due to the sublimation of sulfur is found. It would take a couple of years to sublimate a 1 cm thick layer of sulfur concrete on the Moon at room temperature, but only 2 h at 120 °C. [46] Using a top layer of loose regolith, the temperature would be more constant in the range of −20 °C or lower without daily changes (at a depth of 30 cm) [115], and this is less detrimental in terms of sulfur sublimation due to the low vacuum. The atmospheric conditions on Mars with a temperature of 35 °C or lower and sufficient pressure will not cause sublimation effects. [48] Due to a difference in gravity of about 3/8 of Earth's, the formation of crystals may also be altered. Examples of such research are found for Portland cement mixtures, studied under microgravity conditions using parabolic flight sequences lasting 20 s each [116,117] or on the ISS space station. [118] Adding other materials influences the mechanical properties. For example, upon adding aluminum fibers to a mixture with 35 % sulfur and Lunar regolith as aggregate type, the compressive strength increased from 34 MPa to 43 MPa (average compressive strength value, original data were not given in the report). [13] The addition of glass fibers, however, reduced the strength. [119,120] Care should also be taken for the possible abrasive dust on the Moon and Mars, using protected machinery, pavements, and anchorage systems. [56] As the structure of sulfur changes from monoclinic sulfur to orthorhombic sulfur upon cooling down, the material shrinks, forming cavities, internal stresses, and cracks. Only sulfur modified by polymerization can be used; a volume change of 7 % leads to stresses and reduces the applicability. [63]) By adding a polymer (dicyclopentadiene, dipentene, olefinic hydrocarbon polymers, phosphoric acid, or other modifiers), the formation of macro-sulfur crystals is prevented, and the coefficient of thermal expansion is lowered. [42,51,67] Vlahovic et al. [42] tested the durability of sulfur concrete in different aggressive environments. While using different additions such as micro silica, fly ash and aluminium, the research revealed that sulfur concrete modified with fly ash showed better tolerance to acid solutions, while that with aluminium or silica showed better durability in saline solution, and the loss of compressive strength does not depend on the types of aggregates, but on the porosity of the samples. [42] Mohammed et al. [121] also investigated the effect of different aggregate types on sulfur concrete by replacing parts of the coarse aggregate with river sand, manufactured sand, beach sand, and ecological sand in different percentages. The results revealed that the aggregate type does not affect the properties of sulfur concrete. [121] In general, sulfur concrete has higher durability in aggressive (acid and salt) environments. Wrzesinski et al. [108] studied the corrosion resistance of reinforced sulfur concrete, the results indicating that sulfur concrete

TABLE 3

Characteristics of sulfur concrete with different constituents, with the mean and the standard variation in between brackets, where possible. [52]·

Reference	Components (wt. %)	Compressive strength (MPa)	Durability Mass loss (%)	Permeability (md)	Young's Modulus (GPa)
[125]	Quartz (80–50); Sulfur (20–50)	3.5 (±0.4) ∼ 13.5 (±0.4)	–	0.8 ∼ 46 (±0.5)	1 (±0.2) ∼ 12.5 (±0.2)
[98]	Coarse aggregate (42); Natural sand (40); Sulphur pellets (11.5); Modified sulfur (1.2); Fly ash (5.3)	29.5(±5.3)	–	–	-
[42]	Sand (63); Fillers (7); Sulfur (30)	55 (-)	0 (20 % H_2SO_4) −1 (10 % HCl) 2 (3 % NaCl)	–	-
[126]	Sulfur cement (28); Coquina (67.5); Filler (4.5)	45(±4.0)	"Loss in uniaxial compressive strength" 4.3 (H_2SO_4) 3.0 (Na_2SO_4) 2.4 ($MgSO_4$)	–	16.23(-)
[127]	Silica (49); Iron Oxide (11); Aluminum Oxide (17); Calcium Oxide (10); Magnesium Oxide (6); Sulfates (<1); Other (6)	41(-)	–	–	-
[128]	Aggregates (68); Plasticizer (2); Sulfur (30)	37.6(±0.9)	–	–	-
[48]	Sulfur (50); Martian soil simulant JSC-1A (50)	48(-)	–	–	-
[41]	Sulfur (35) Moon regolith simulant JSC-1 (65)	33.8(-)			

has a higher corrosion resistance capacity than conventional concrete because it improves resistance to chloride and sulfate ion permeability and is fully compatible with uncoated reinforcing steel bars. In addition, Ksiazek [122] revealed that a polymer-sulfur coating can reduce corrosion of steel reinforcement under tension; the optimum thickness of this coating is 1.4–1.6 mm, but it is not applicable in the presence of strong oxidizing acids and alkalis.

According to the article by Loov et al. [71]), sulfur concrete has a higher creep than Portland cement concrete. After one month, the creep of sulfur concrete with 25 % sulfur by weight is 3 times greater than the initial deformation. [71] However, according to the publication by El Gamal et al. [123], the compressive creep of sulfur concrete is lower than that of Portland cement concrete. The results are not consistent. Fediuk et al. [52] point out that formulation and application conditions determine low-temperature creep characteristics. Further research is therefore needed to better understand the creep characteristics of sulfur concrete under different conditions. After curing of sulfur concrete, the thermal shrinkage index is not significant; significant deformations are due to temperature variation, which is proportional to the coefficient of linear thermal expansion. [52].

El Gamal et al. [124] tested the effects of the wastewater environment on Portland cement concrete (PCC), sulfur-resistant cement concrete (SRC) and modified sulfur concrete (MSC). Alkalinity tests on PCC and SRC showed neutralization at depths of 3.8 mm and 3.2 mm respectively. MSC, on the other hand, showed no change in alkalinity. Furthermore, the pores of MSC did not change significantly, whereas PCC and SRC showed an increase in the number of pores, as well as in the connectivity between pores.

Previous research has established a general knowledge frame about the properties of sulfur concrete. Table 3 provides an overview of research results about the effects on the properties of sulfur introduced by different component types and content.

Further investigation found that sulfur concrete showed shrinkage cracking, which leads to premature failure. Evidence explained that it is due to the transformation of sulfur from β (rhombic) to α (monoclinic), which leads to shrinkage of up to 7 %, introducing high interstress within the mixture. [42,123] Starting from the 1970 s, after plenty of work, researchers insisted that unmodified sulfur concrete could not replace Portland cement concrete. [129,130,71] Hence, modified sulfur concrete became the topic of further investigation to optimize sulfur concrete.

The trials for modification of sulfur concrete started with the aggregates rather than chemical additives: Gregor [131] concluded that compared with the spherical gravel, the crushed stone is more difficult to impact, which leads to higher porosity; thus, 10–20 % more sulfur is needed as the binder. Meanwhile, when the aggregate distribution is the same, sulfur concrete with crushed stone has 32 % higher compressive strength than the one with spherical gravel. [131] Besides, previous research shows that limestone-formed sulfur concrete exhibits higher strength than the one formed with silica, which has lower strength. [131].

Except for the variation of the content of sulfur change in types of aggregates, the application of organic additives, such as dicyclopentadiene (DCP), is crucial progress to modify the properties of sulfur as the binder followed as the most utilized method for obtaining modified sulfur concrete. McBee and Sullivan [79] prepared modified sulfur by adding 5 wt% DCP in small increments respectively, reacting with sulfur, then utilized limestone and silicon as the primary aggregates to fabricate modified sulfur concretes. The mechanical properties, freeze-thawing resistance, acid corrosion resistance, and salt corrosion resistance were tested and discussed. Based on these discussions, the advantages of modified sulfur concrete by DCP are given as follows: [79].

- In most cases, the modified sulfur concrete had superior mechanical properties (62.2 MPa for compressive stress, 6.2 MPa for tensile strength, 9.2 MPa for flexural strength, on average) comparable to high-strength Portland cement concretes, especially for limestone-modified sulfur concrete;
- The residual flexural strength of modified sulfur concrete is around two times that of the unmodified sulfur concrete after 300 times freeze–thaw cycles;
- Exposure of modified sulfur concrete to acid solutions for 1 year caused negligible corrosion or erosion, and the mechanical properties remained;
- As in acid resistance tests, no measurable attack on the modified sulfur concrete specimens upon exposure for 1 year to the salt solutions was found;
- After 1 year of immersion, water penetration into modified sulfur concrete is negligible, and the material is impermeable.

Researchers further developed modified sulfur concrete with different additives and methods based on their work. Bae et al. [100] used 15 wt% of modified sulfur mixed with 13 wt% of fly ash, 32 wt% of sand, and 40 wt% of coarse aggregate; the obtained modified sulfur concrete gave 83 MPa of compressive strength, 13 MPa of flexural strength, and 6 MPa of tensile strength, on average. Anyszka et al. [99] found that the compressive strength of modified sulfur concrete by 5 wt% DCPD and 5 wt% styrene even increased more than 600 N after 5 times melt recycling. Besides, the resistance of sulfur concrete to the aggressive environment (water and temperature cycles, H2SO4, HCl, and NaCl) is better than cement concrete. [51,132–134] Following the information in Table 3; sulfur concrete is significantly more resistant to acid attacks than cement concrete. Still, it is less resistant to microbial corrosion, especially sulfur-oxidizing bacteria. [132].

Overall, sulfur concrete has exhibited many aspects of better performance than cement concrete, including mechanical properties, fast early strength formation, and better resistance under aggressive environments such as exposure to acids. Still, the internal stress and cracks induced by a transformation of sulfur allotropes' loss of integrity during heating limit its application. The motivation to replace cement concrete with sulfur concrete is sustainability, linked to decreased CO_2 emissions and less water usage. The increasing surplus of sulfur from the modern petroleum industry offers plenty of raw materials for sulfur concrete. Furthermore, sulfur concrete does not need water to be fabricated, while potable water will soon become scarce. Those two motivations match well with the futural development of human society, where we need to build in a better, more robust, and more sustainable way for future generations. It is worth noting that the broad application of sulfur concrete may be achieved far from the Earth.

Building on the Moon and Mars, their environment and in-situ resource utilization (ISRU)

As mentioned in the introduction of Chapter 2, it has been around half a century since sulfur concrete was first proposed as a possible construction material for the extraterrestrial environment. It is necessary to build reliable, sustainable, and permanent buildings on the Moon to execute in-situ exploration on the lunar surface and even further permanent habitats on Mars. First, the characteristics of the extraterrestrial planet's impact on the constructions should be investigated to achieve the aim of being able to exist in an extraterrestrial environment. Besides, to avoid a tremendous budget for transferring construction materials from the Earth to other celestial objects, exploiting and utilizing local materials should be the first idea of searching for resources, especially the materials for fabricating sulfur concrete.

Lunar environment

The Moon is the nearest and permanent satellite of Earth. However, according to the lunar probes, the environmental conditions of the Moon are much harsher than those on Earth, at least in terms of being a habitat for humans. The first consideration is the thinner atmosphere of the Moon, where the galaxy's cosmic radiation readily punctuates through, which is unacceptable for both human and electronic devices. Furthermore, the lunar craters prove that without a thick atmosphere, even micrometeorites with mass less than 10^{-2} g at a speed of 20–70 $km \cdot s^{-1}$ would crash on the surface of the Moon, which is unacceptable for both buildings and humans. [41] Toutanji et al. [135] tested and simulated the influence of hypervelocity impacts on a sulfur concrete structure, indicating that a 20-year interval of serious damage by meteoroids is fatal to manned structures. The second consideration is the huge thermal gap found on the Moon, which varies from –233 (lowest temperature at poles) to 123 (highest temperature at equator). [115,136,137,41,43] The mean surface temperature is 107 °C during sun exposure and −153 °C during dark condition. Earth, having an average temperature of 22 °C, has lesser extremes, ranging from −89 °C to 58 °C. [138] The thermal differences found on the Moon mean that future construction materials will have to offer superior performance in terms of thermal expansion and shrinkage. [41,138] The lunar gravity is approximately one-sixth of that of Earth. [41,138] Meanwhile, the advantages are also favorable for construction, such as the low gravity, which leads to much less self-load of construction materials, and much lower seismic energy, which can almost be ignored compared to Earth. [41,48,139].

Lunar concrete or lunarcrete

The surface of the Moon is covered by so-called lunar regolith. Compared with Earth, no evidence has been found of the existence of a flow of water, oxygen, or other conditions necessary for life activities on the Moon. The lunar surface is affected by possible meteorite activity, the thermal gap, the bombardment of cosmic rays, and solar winds. Thus, the top layers of the regolith are always loose. [138,140,141,2] Even with much lower gravity than Earth, an adult can leave footprints on the surface of regolith, and by the depth of 30 cm, the lunar soil becomes compacted. [142,143].

From a view of the granulometry of lunar soil, the lunar regolith has grain sizes from 46 μm to 110 μm, [144], and the average particle size is between 60 and 80 μm (Fig. 1a). [2,142] From a view of chemical composition, half of the weight of regolith is a composite of SiO2, TiO_2, and other metallic oxides. [145] In general, lunar regolith has the potential to be used as filler or aggregates

(a) (b)

FIG. 1

Simulated lunar regolith sample [151] **(a)** and Lunarcrete [147] **(b)**.

for concrete, [43,146] Besides, because the main component of lunar regolith is aluminosilicate, which contains glassy phases and is ideal for reacting with alkali activators, the lunar regolith also has huge potential for being used as a geopolymer material. [147] Due to the limited lunar regolith amount on Earth, currently around 400 kg, [15] researchers focus more on simulants mimicking lunar regolith. The lunar simulants are made from terrestrial materials, aiming to obtain lunar-regolith-like soil with chemical compositions and particle sizes as close to the actual lunar regolith. [11] Chen et al. [148] treated lunar soil simulant mixed with 2–5 wt% polymer binder under high-pressure compaction. The consequent inorganic–organic monolith exhibited high flexural strength around 30–40 MPa. Based on their results, Su et al. [149] confirmed that this concrete's fatigue behavior is better than cement concrete's. Oh et al. [150] mixed JSC-1A lunar soil simulant with 4 wt% epoxy binder, and after dropping a hammer on the premixed simulant and binder, the flexural strength increased with the impact energy. Without a doubt, it is a popular trend to obtain a reliable lunar construction material by mixing it with another optimizer, changing the curing conditions, or inducing external influence (Fig. 1b).

Environment of Mars

Compared to the Moon, the environment of Mars is more friendly to humans: the rotation period is quite similar to Earth, which is 1.026 Earth days, and the surface gravity is around one-third of Earth. [2]) Besides, the surface temperature of Mars is around −140 to 20 . [152] In addition, Mars possesses an atmosphere similar to that of the Earth, [153] thus, the objectives on the surface are protected from cosmic radiation and meteorites. It is known that dust lifting and sand transport by wind-dominating geological activity exist on Mars. [154] Surprisingly, in 1972, after the arrival of the Mariner 9 spacecraft, recordings were taken showing sinuous channels and valleys on the surface of Mars. [155] Additionally, the Viking missions by NASA obtained more images of valleys and channels in higher resolution. In 1998, the Mars Global Surveyor captured an image of the "taffy-pull" terrain of northwestern Hellas Planitia on the surface of Mars; it is inferred that the formation of this terrain is related to the erosion of different layers of bedrock and mass movement of debris. It also shows evidence for using water on Mars, [155–158] which allows the production of cement concrete. Still, it is a challenge for all the explorers on Mars to isolate available ice deposits. [159] Surprisingly, the Chinese Martian rover found liquid water on Mars. [160].

Nevertheless, Mars is regarded as a sulfur-rich planet. The sulfur cycle is the most important geochemical cycle on Mars, and countless sulfates and sulfites exist on the surface of Mars, in amounts much higher than those on the Earth, the Moon, and asteroids. [161,162,49] Previous research has analyzed the sulfur elemental abundance from both in-situ landers and remote sensors on the surface of Mars and from Mars meteorites. The results indicated a broad range of sulfur content on Mars, which amounts to 2.2 wt% for the bulk of Mars, [163] and from 5 to 10 wt% [164] or even more than 20 wt% [165] for the core of Mars. It is promising to expect abundant sulfur resources and a recyclable, sustainable, and durable application for sulfur-related materials on Mars. In general, with all the evidence found above, researchers insist that Mars is the most habitant planet in our solar system for human beings except for Earth. [2,48,166–168].

Mars concrete or marscrete

Since favorable living conditions are found on Mars, the exploration of creating buildings on Mars never stops. In 1996, McKay first proposed that concrete is preferable for construction use on Mars. [169] Specifically, Martian concrete composed of Martian regolith and related simulants (e.g., JSC Mars-1A (Fig. 2a) [19] has been investigated comprehensively. Wan et al. [48] fabricated mixtures using maximal aggregate sizes of 1 mm and 5 mm of JSC Mars-1A simulant with different percentages of sulfur (Fig. 2b). It can be observed that the mixture with smaller aggregates resulted in a smoother surface. In contrast, the larger aggregates gave uneven, rough surfaces with many voids distributed around larger aggregates. Different composite designs related to Martian sulfur-based concrete are proposed, except for the relatively simple composition of sulfur and aggregates, as shown in Table 3. Li et al. [170] mixed magnetite and silica sand into powder sulfur, aimed at a better combination of binder and aggregates during fabrication.

Sulfur concrete can be made in different ways. One example is the casting process using microwaves. Components, including

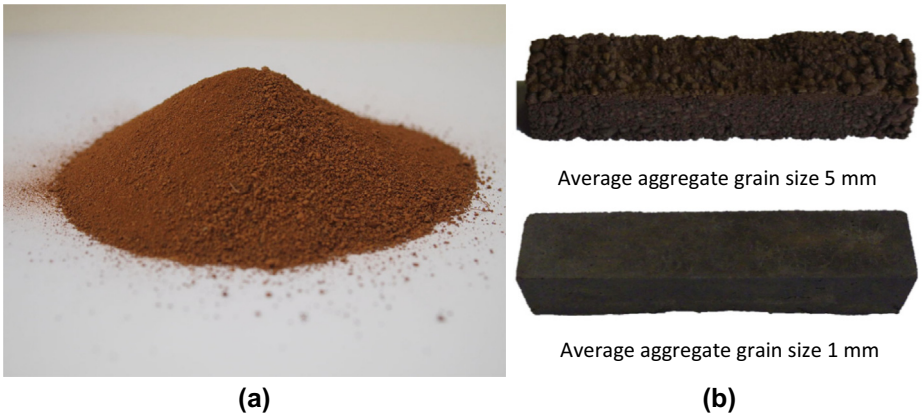

FIG. 2

Martian regolith simulant JSC Mars-1A developed by NASA [171] **(a)** and Martian concrete samples composed of JSC Mars-1A with a maximum of 5 mm aggregates and 1 mm aggregates [48] **(b)**.

sulfur powder, magnetite aggregates, and silica sand, are mixed uniformly inside a beaker and then put into the microwave oven. The ingredients are heated to setting temperature and then moved out. After cooling down for around 24 h, the fabricated specimen can be smoothly removed from the beaker, as can be seen in Fig. 3. [170].

As for the exploration for better performance of Martian sulfur-based concrete, Scott et al. [172] fabricated and tested the compressive strength of a mixture composed of Mars regolith simulant, magnesium oxide, amorphous silica, and water, being 35 MPa after 90 days. It is notable that due to the environmental conditions of Mars, some construction materials unsuitable for construction on Earth become potentially helpful, such as the Plaster of Paris, which is obtained from the calcining of gypsum around 150-165 . [173] The presence of water inside the gypsum impacts the reinforcement of concrete and the gypsum itself. [173,174] Reches [175,176] concludes that the water is not thermodynamically stable on Mars, which gives the possibility for Plaster of Paris being used as a binder for concrete because all the water in the pores of gypsum is evaporated or frozen. [175,176] However, water use should be avoided as much as possible.

3D printing technology is also proposed to produce base structures combined with in-situ resources for Mars. Kading and Straub [177] summarized the idea of in-situ construction building on Lunar or Mars into two stages. First, unmanned space crafts land on the surface to print the main domes using inflatable structures. The process can be seen in Fig. 4. [61].

In a subsequent stage, manned space crafts would send astronauts to where the infrastructures are built during the initial phase, continuing to finish works that the machines cannot accomplish automatically. The methodology of Fused Deposition Modeling (FDM) is proposed in common use for basalt 3D printing on Mars, [178] because the content of basalt is higher than any other planet in the solar system. [179] Besides, Wang et al. [180] summarized other additive manufacturing techniques potentially accessible for Lunar/Mars concrete, including direct ink writing, [181] powder-based ink jetting (see Fig. 5), and [182] powder-based selective laser melting. [183] Generally, using sulfur concrete on Mars seems more feasible than on the

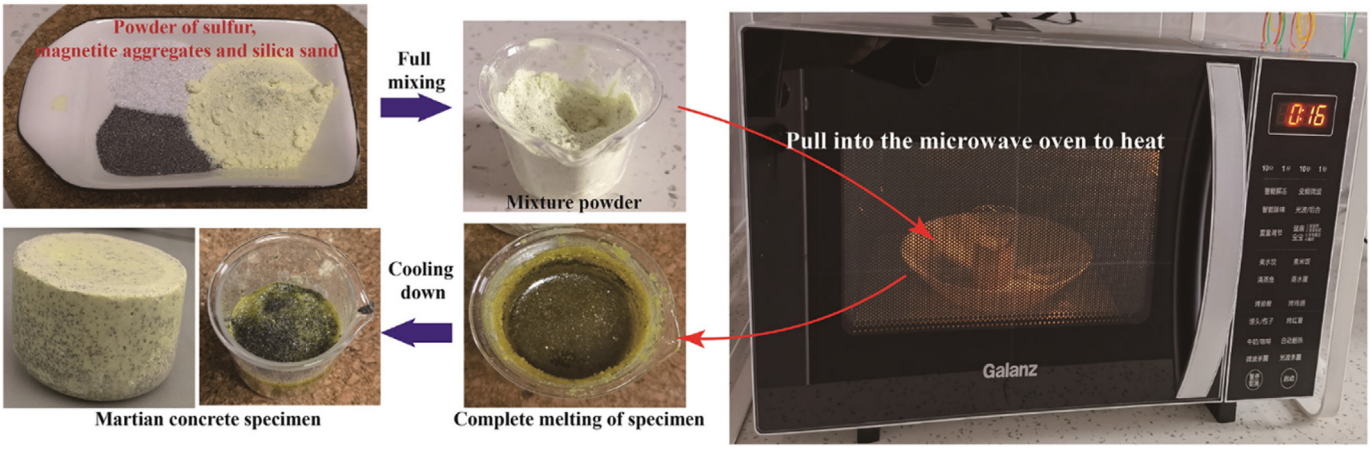

FIG. 3

The casting process of Martian concrete by microwave. [170].

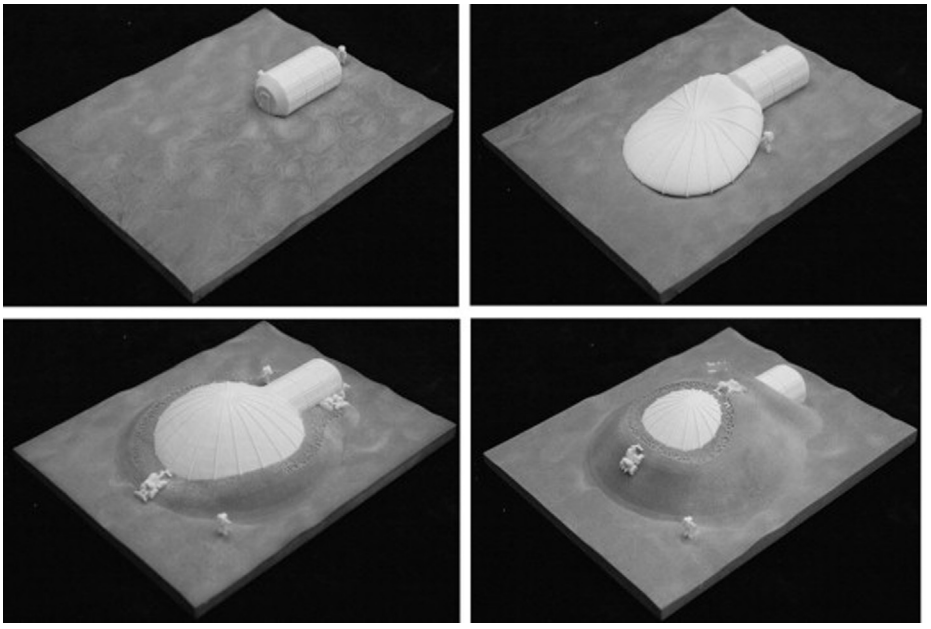

FIG. 4

Exemplary sequence of the lunar outpost construction by means of two cooperating autonomous vehicles. [61].

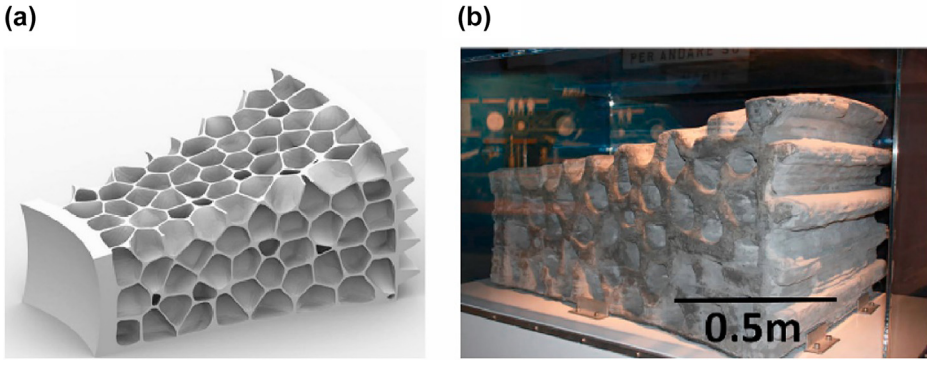

FIG. 5

CAD model (a) and printed structural element technical demonstrator-2 (b) [182].

Moon. Still, exploring other aspects of those customized construction materials is necessary, not only focusing on mechanical properties.

As mentioned before, Mars is a sulfur-rich planet; the same methodology is possible for using in-situ sulfur resources to fabricate sulfur concrete.

ISRU on an extraterrestrial celestial object

As emphasized before, the cost of transporting raw materials from the Earth to the extraterrestrial planets, including the Moon and Mars, is unacceptable for economic considerations. According to the history of human development during and after colonization, making full use of the local resources should be the first idea of each exploration to decrease the cost and increase the compatibility of new materials and their derivatives.

ISRU refers to the in-situ generation of consumables from raw materials found on other planetary bodies for autonomous or human activities. [184–186] The necessity of ISRU is mainly about a significant reduction in the budget of materials transportation between Earth and other celestial objects. [137,187,188] Crawford [142] proposed a diagram to describe the energetic consumption of vehicles between Earth and the Moon, where the vertical axis represents the energy needed to reach a stable orbit from a given distance from the Earth, converting into the total amount of height to reach by overcoming a constant Earth gravity. If one attempts to send materials from the surface of Earth, the energy corresponding to climbing 6400 km under Earth's gravity is needed. In comparison, it is only around 250 km from the Lunar surface. It is the same for Mars, and according to previous research, each kilogram of useful technology sent to Mars needs 5.6–8.8 kg of fuel, [189] even one attempt to build up a local fuel depot on Mars, 226 kg of propellant is required to obtain 1 kg of fuel on Mars. [190] NASA conducted several ISRU programs for Mars, including water, oxygen, methane, [191] robotic construction [192], and metals, silicon, and other feedstock commodities. [193].

The extraterrestrial resources are expected to be applicable generally in three aspects: 1) as complementary or commercial resources on Earth; 2) as a contribution to the production of extraterrestrial materials for local basement use; and 3) as further utilization of exploration of other celestial objects. So far, Lunar and Martian regolith analysis exhibits potential valuable resources such as metal, oxides, glass, and sulfur. [11,194,195] The moon formed after collisions between a Mars-sized impactor and the Earth, as debris ejected into the universe and coalesced. [12] Thus, the volatile elements such as sodium, potassium, and water of lunar rocks are much less than the components of terrestrial rocks due to the lack of atmosphere. [138] The majority of a constituent of lunar regolith is similar to the constituent of the Earth's crust, which is composed of 40–50 % SiO2 and 10–20 % Al2O3. [147] The knowledge about Martian regolith, however, is only obtained from robotic exploration missions since the samples are so far not available. NASA will obtain two new regolith samples of Mars (broken rock and dust) in the near future. [196] Due to the rare Lunar/Martian regolith, researchers mainly focus on extraterrestrial regolith simulants and related ISRU methodologies. Karl et al. [197] summarized previous Martian regolith simulants into three generations, which indicated that the development of Martian regolith simulants was mainly about the complexity and physical properties of components. A schematic diagram of the development of Martian simulants is given in Fig. 6.

Meanwhile, practical processing methodologies on the Moon and Mars are being proposed, such as melting and sintering (Fig. 7). Dry-Mix/Steam Injection (DMSI) and other methods are also being developed for better application of ISRU. [12,41,198–202].

Barkatt and Okutsu [49] propose different approaches for obtaining elemental sulfur on celestial objects, and especially on Mars:

- Chemical oxidation or decomposition of sulfides: compared with chemical oxidation, thermal decomposition of sulfides does not require an oxidizing agent and does not produce by-products such as SO_2 or SO_3.
- Electrochemical oxidation or decomposition of molten sulfides: it is necessary to distinguish sulfide minerals that do not require high temperatures for electrolysis, using solar energy.
- Chemical reduction of sulfates to elemental sulfur: using CO or methane as a reducing agent: these agents exist on Mars, and the treatment approach has been successfully carried out on previous space missions.
- Decomposition of sulfates in the molten state using electrochemical methods: the main obstacle is that the possibility of obtaining sulfites is higher than that of obtaining elemental sulfur, but the by-product can be used as raw material for further processing.

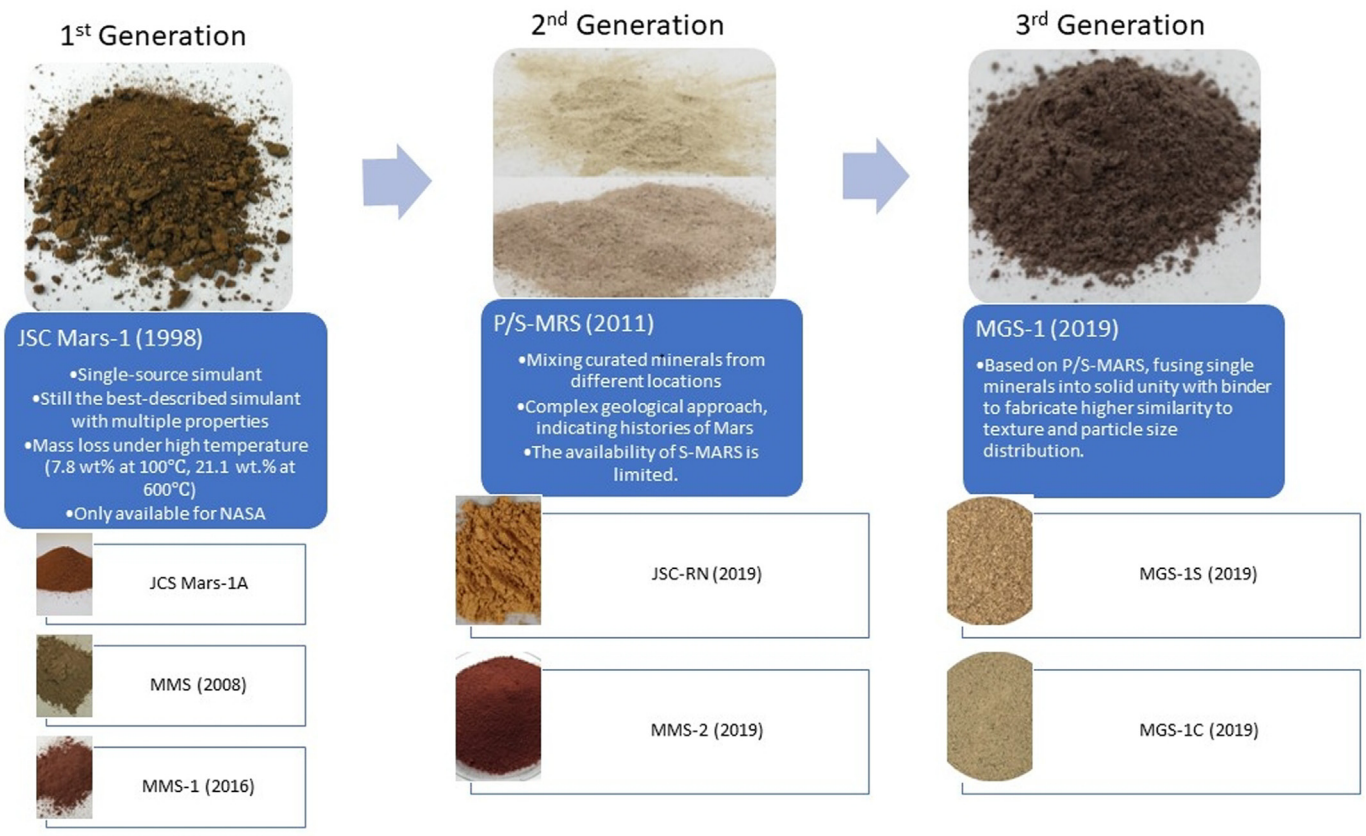

FIG. 6

Schematic picture of the development of Martian regolith simulants. [197].

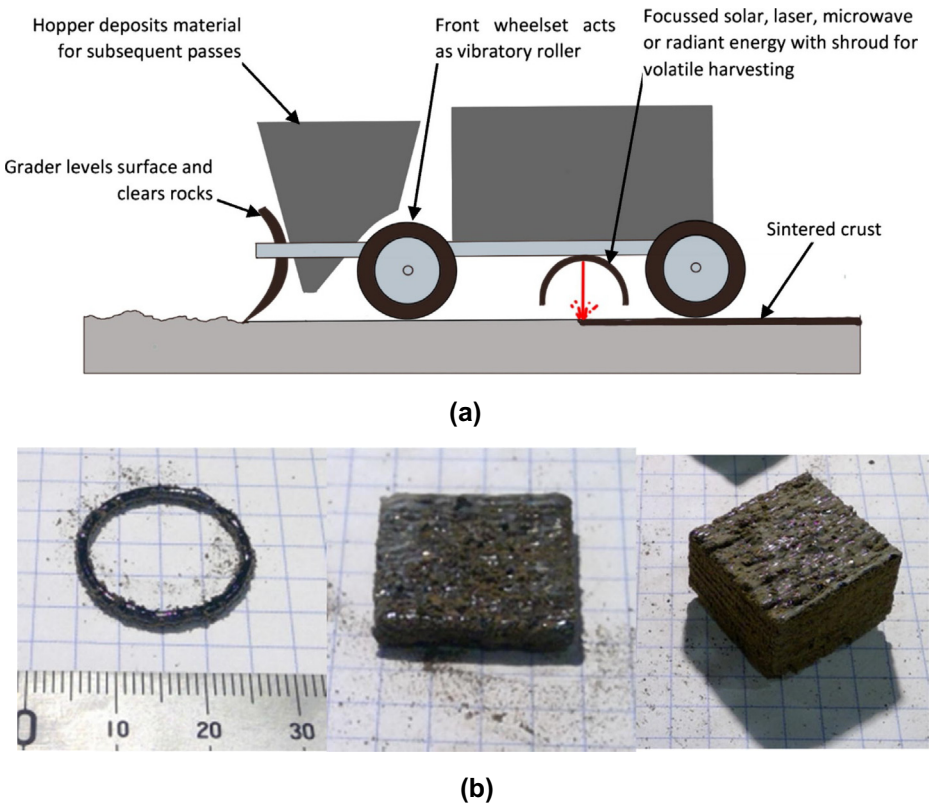

FIG. 7

Autonomous vehicle concept for direct sintering of regolith (a) and objective produced by selective laser melting process (b) [12].

These methods can also be used to obtain the element sulfur on the Moon and still need to be studied in more detail. However, at the time of writing, the Indian Chandrayaan-3 lander detected sulfur near the Moon's poles, meaning that the Moon's sulfur content could be higher than previously thought, further increasing the accessibility of sufficient elemental sulfur on the Moon. [203].

Specifically, Yuan et al. [204] explored the contour crafting performance of sulfur cement mixtures. Fig. 8 shows the problem of ununiform sulfur melting during heating, leading to a difference between external and internal rings. This also leads to crack formation. Pre-heating of Martian regolith simulant is vital to ensure the uniformity of sulfur concrete throughout the sample. However, this research shows the possibility of using contour crafting techniques to construct sulfur concrete structures.

Sulfur concrete: Preliminary exploration for a sustainable building material

As a summary of previous chapters, sulfur concrete's application in terrestrial and extraterrestrial environments is limited mainly due to the following aspects. First, transforming the sulfur element under different temperatures leads to internal stresses and, consequently, internal cracks. Secondly, Grugel [47] found different thermal expansion coefficients between sulfur ($6.4 \cdot 10^{-5}$ K^{-1}) and aggregates ($5.5 \cdot 10^{-7}$ K^{-1}) for silica, leading to failure at 7 MPa in compression compared to 35 MPa for reference specimens, in temperature cycles between −191 °C and 20 °C, due to ununiform expansion leading to debonding inside the sulfur concrete under thermal differences. Thirdly, in terms of extraterrestrial colonization, the vacuum environment significantly increases the speed of sublimation of sulfur from sulfur concrete,

FIG. 8

Extruded sample with cracks on surface (a) Extruded sample after processing of pre-heating of Martian regolith simulant sulfur concrete, a smooth surface is achieved (b) [204].

which can sublimate a 1 cm thick layer of concrete under 120 within 2 h. [46] improvements to optimizing the abovementioned disadvantage should be conducted for a more comprehensive terrestrial application and even further application under extraterrestrial environments.

Current research for better performance of sulfur concrete

Research has revealed that the aggregates of sulfur concrete not only offer strength by bonding but also work as a modifier during the cooling process, which prevents the formation of sizeable cyclic sulfur crystals, thus securing the stability of mechanical properties. [42,63,205]) Besides, solid sulfur plays two roles in terms of structure: one is to work as a binder binding aggregates by physicochemical actions, while another one is to provide structural strength together with aggregates since the crystal-modified parts of sulfur obtained higher modulus than the parts working as the binder. [206,207,47,52] Except for sulfur, external modifiers [76,205,207–211] are added for better microstructure by controlling macro-crystals' crystallization. Recently, Mohamed and El Gamal [123,124,205,212–214] contributed to the exploration of sulfur polymer concrete, primarily related to the durability of modified sulfur concrete under an aggressive environment, including an actual sewage environment. The modified sulfur concrete showed a much lower weight loss of only 2.2 %, while Portland cement concrete and sulfate-resistant concrete showed a loss of 8.0 % and 13.6 %, respectively. [212] A new sulfur polymer concrete was fabricated by mixing modified and elemental sulfur, fly ash, and desert sand by Mohamed and Gamal. [205] This newly developed sulfur polymer concrete exhibited high compressive strength, low hydraulic conductivity, and particularly corrosion resistance under acid and salt environments. [205] Furthermore, they also explored using recycled aggregate materials to fabricate modified sulfur concrete to use sulfur-based hazardous waste sustainably; the product can be used to retain structures, foundations, road constructions, etc. [213] It can be concluded that dicyclopentadiene and olefinic polysulfide are the most popular additives. However, the former reacts with sulfur exothermally. The temperature control during the reaction is hardly feasible due to the complexity, and the modified sulfur is unstable when exposed to high temperatures.

It is also important to look further into the question of obtaining elemental sulfur on the Moon and Mars. The techniques mentioned in section 3 may prove feasible, and actual tests should be carried out in the future. As sulfur is present in the outer layers of the surface, elemental sulfur will be obtained. On celestial objects, the obtained sulfur can also be heated, for example using solar heating or laser technology.

Phase-change materials: A solution to improve the thermal performance of sulfur concrete

There is no doubt that the temperature should also be controlled in care for better stability of sulfur concrete; one of the logical ways is to prevent the interaction between sulfur concrete and extra thermal difference, especially when applying sulfur concrete for real-life utilization. Phase-change materials (PCMs), such as paraffin (derivates), could be promising for this attempt. In general, the phase change materials have high latent heat, indicating a large amount of absorption/relaxation of heat when the phase changes (solidification/melting) happen, thus decreasing the temperature difference. [215]) PCMs can be divided into three classifications: organic, inorganic, and eutectic. [216] The applications of PCMs are primarily related to building energy comfort. [216–219] The idea is to use PCMs to optimize the heat capacity of conventional concrete by full utilization of the absorption, relaxation, and storage abilities of the heat of PCMs, thus minimizing the temperature changes of concrete to avoid thermal cracks, and save energy. Usually, there are three basic approaches to combining PCMs with concrete: 1) replacement of part of sand in concrete by encapsulated PCMs, 2) direct addition of PCMs into concrete when mixing the mortar, and 3) impregnating porous materials with PCMs. [218] So far, a wide temperature range from −20 to 200 of PCMs in the application for heating, cooling, or combined heating & cooling progress are used. [220] Using PCMs, the cementitious material's hydration peak temperature and thermal comfort are influenced. However, the compressive strength was also lowered with increasing amounts of PCMs. [215] Furthermore, PCMs should be limited within the range of 3 wt% of binder to ensure an acceptable reduction of compressive strength; other researchers also reported the same problem. [221,222] Nano silica and carbon fibers are proposed to counteract these issues, next to other alternatives. [223–225].

Exposure to temperatures close to the melting point leads to problems with the integrity of sulfur concrete. So far, the application of PCMs to conventional concrete has been verified. Liu et al. [226] presented the thermal performance of a new PCM-based concrete in four different cities. The results show that PCMs significantly increase the thermal storage capacity of concrete while remaining within the optimum temperature range. Based on the research carried out on conventional concrete, an analogy can be proposed for sulfur concrete to further reduce the effect on its application due to the temperature on the Moon, which can be higher than the melting point of sulfur. The compatibility between PCMs and sulfur concrete is still uncertain. However, the same approach as a PCM foam panel [227] could be applied to sulfur concrete, thus dividing the two materials and avoiding potential compatibility problems. At present, the application of sulfur concrete remains limited and mainly concerns underground applications such as sewer systems. However, it is also possible to use it as a sleeper in railway applications. Further research could prove the applicability of sulfur concrete in other applications. For example, some structural elements may be exposed to daily solar radiation, which raises surface temperatures up to 65–70 °C. [228,229] This phenomenon can be countered by the use of PCMs, as mentioned above. [230] Even though this temperature is below the melting temperature range of sulfur concrete, it should be avoided, especially for load-bearing elements. However, if direct sunlight can be avoided with a finishing coating, this problem is avoided.

Self-healing: A solution for early-age and long-term cracking of sulfur concrete

One of the most crucial problems for concrete structures is cracks introduced by thermal gaps, vibration, corrosion, fatigue, etc. Though engineers and researchers tried different approaches to eliminate the appearance of crevices, cracks smaller than

300 μm are almost unavoidable. [231] Sulfur concrete is a typical example of this issue: complex thermal control systems make it economically unacceptable if one attempts to eliminate early-age gaps when sulfur concrete is curing. Furthermore, it is likely that during the service of sulfur concrete, plenty of other reasons would cause cracks inside or outside the sulfur concrete because of the sensitivity of sulfur concrete to temperature differences. It would be difficult and expensive to detect and repair some of the cracks, which significantly impact sulfur concrete's mechanical properties and durability or at least accelerate its degradation. Recently, researchers found that employing self-healing, which is feasible for healing micro-cracks, will potentially achieve more durable and cost-effective construction elements.

Concrete, as the construction material possibly lasting for thousands of years, has the characteristic of autogenous self-healing, which indicates no treatment shall be made, and the concrete will heal the cracks by itself. Most of the contribution belongs to the cement that is not hydrated during curing, and calcium ions leaking from the concrete and carbon ions forming precipitation. Those actions create a new component for filling cracks. [231,232]) Different methods are addressed to optimize the self-healing of concrete, including the addition of mineral components such as supplementary cementitious materials, [233] superabsorbent polymers (SAP), [234] polymers, [235] expanding agent, [236] bacteria, [237] vascular healing concrete [238] or other encapsulated additions. [239] Research about self-healing related to sulfur composites has not yet investigated in detail. Wan et al. [48] tested a mixture of various percentages of sulfur with JSC Mars-1A regolith and conducted an unconfined compression test and a three-point bending fracture test. The results indicated that the recast of Martian Concrete can further increase the material's overall performance. Gwon et al. [240] used calcium sulfoaluminate (CSA)-based cement with SAP, mixed with modified sulfur and finely graded aggregates to fabricate sulfur composites. They focused on improving the durability of cracked sulfur composites by crack-sealing. The results showed that modified sulfur concrete without SAP and CSA binary cement has significant crack-healing, with a 43 % reduction in effective crack width after 7 days of water immersion. Meanwhile, Snoeck et al. showed that adding SAP increases the rate of autogenous self-healing, which is confirmed in cementitious systems up to an actual age of 10 years, with a partially repeatable healing possibility. [241–244] Gulzar [95] investigated the mechanical properties and durability of the sulfur concrete after 1 and 2 times recasting; the results indicated that the compressive strength increased for the first time recast. However, a significant decrease appears after the second time recast. Hence, many questions remain about the recyclability and linked mechanical performances of sulfur concrete, information which is necessary to investigate its self-healing potential in the future.

The observed cracking in sulfur concrete causes durability issues and a loss in mechanical properties. Current sulfur concrete elements are susceptible to cracking and require maintenance and repair. However, not all structures and components are easy to access for inspection and repair, and it is costly to repair structures on celestial objects. Employing self-healing, more durable and sustainable structures can be designed and constructed. This will reduce the maintenance, inspection, and repair costs, leading to a lower life cycle cost. [245] Since sulfur is a thermoplastic material, an analogy can be made to asphalt, which can repair its cracks and recover its mechanical properties and fatigue life during rest periods. Asphalt, the most frequently used material in pavement engineering, shows a temperature-related healing phenomenon after repeated loadings. [246] Bitumen can be healed using different mechanisms, ranging from inductive heating with a high-frequency alternating electromagnetic field or microwaves to embedded rejuvenation encapsulation dissolving the bitumen. [247–249] The temperature is the most determining factor and a high temperature increases the self-healing rate and shortens the time needed for complete healing. [250,251] Using high-frequency alternating electromagnetic fields, eddy currents can be induced in electrically and magnetically susceptible materials. Steel fibers can be added to asphalt mixtures to increase the healing capacity with inductive heating when cracks occur in the asphalt mastic. [252–254] The metallic threads are heated, and this heat energy diffuses into the bitumen to increase the temperature. Long steel fibers are more effective for conducting heat but are very difficult to mix in as they tend to form clusters, decreasing the mechanical properties. [255] The healing of asphalt mixture is due to the thermally activated process of capillary flow and diffusion of the binder, and the healing is around 78 % when heated up to 85 °C. [256] Not the complete capacity of healing can be used, and this may be due to broken aggregates or the inhomogeneous temperature gradient in the depth of the asphalt. Microwave heating is also used to induce self-healing. Due to an alternating magnetic field, the orientation of polar molecules changes, increasing the temperature. [254,257] Different properties affect the healing, including the bitumen and asphalt mixture characteristics, modifiers, temperature, loading condition, rest periods, moisture, etc. Furthermore, the inhomogeneity of temperature gradation inside the asphalt will also lead to an ununiform distribution of temperature, and ununiform healing would introduce internal stresses, reducing durability. [252,255,258] The basic idea of the rejuvenator of asphalt is to diffuse it into the aged binder and restore its original molecular structure to extend the pavement service life. [259,260] Another pathway of using capsules also proved helpful in self-healing asphalt materials. [261,262] In general, the previous research about the self-healing of asphalt materials inspires the self-healing of sulfur-based concrete materials, especially for selecting self-healing methodologies.

There is undoubtedly a need for intelligent materials with self-x capabilities for use in smart cities. [263] A self-healing approach is discussed above, but when looking at self-healing structures, healing is most efficient before the final extended cracking and ultimate failure. There is a need for a continuous healing performance to ensure the structure's health and to repair microcracking, which is not visible by the naked eye (and cannot be visually monitored when living or working on a celestial object). An adequately designed material will lead to safe, long-lasting, and optimally functioning infrastructures, reducing life-cycle costs, less resource consumption, and causing less environmental pollution.

Conclusions and further research work

This paper reviewed sulfur concrete as a material for the Earth, the moon, and Mars. The following conclusions can be drawn.

Though existing for a period, applications of sulfur concrete and the corresponding development of its properties still need to be further enhanced and improved; according to the information from the content of this paper, the necessity, accessibility, and reliability of further research of sulfur concrete are apparent. First, sulfur-based concrete shows the same or even better mechanical properties in terms of application under terrestrial environments. Meanwhile, it has better corrosion resistance under aggressive environments such as water immersion, acid, and salt. It exhibits extended durability, making it a proper construction material for particular parts of civil infrastructure (foundations, ports, sewers, etc.). Another vital motivation is focusing on sustainable development; with the development of the modern petroleum industry, as one of the by-products, sulfur surplus increases rapidly, and the amount of cost for storing or innocent treatment of that heavy surplus would be unimaginable. At the same time, carbon emissions from traditional cement industries also introduce problems in terms of carbon emissions. Combining those two facts, sulfur concrete should play a more critical role than ever, even though the applications remain limited. More importantly, the potential recyclability of sulfur concrete prevents the repeated production of construction materials, leading to less raw materials use reducing the imminent material scarcity, including water as a precious resource. Secondly, in terms of extraterrestrial applications, from the point of view of the economy, it is not cost-effective to transport materials from the Earth to any other extraterrestrial objects, ISRU gives solutions to the need for raw materials for constructing a lunar base or Martian base. Sulfur and its derivatives are available in the Moon regolith, and especially on Mars, the command for the energy of heating sulfur into the liquid phase as a binder of cementitious materials is achievable by utilizing solar energy. The need for no water is always one of the most essential advantages of sulfur concrete. All those advantages combined make sulfur concrete one of the most promising construction materials for future demand.

Still, disadvantages such as poor performance under high temperatures transformation between different phases of sulfur element leading to internal stresses should be solved. In future research, trials inspired by analogies from other construction materials should be the first step, such as applying PCMs in cement concrete for better thermal performance and internal comfort, using metallic fibers in asphalt materials for better self-healing activity, etc.

As a main exciting feature of sulfur concrete, the potential of recyclability and circularity should be further investigated. The link between service life and improving performance after recycling must be clearly defined and linked to the mixture composition. To comprehensively introduce sulfur concrete into the construction industry, numerical models related to its failure mechanisms should also be created to decrease the experimental work and its consumption.

SciVal topics and clusters

For this article, the references come mainly from Scopus and Google Scholar, the search time interval focuses on the last 10 years related to keywords such as sulfur/sulphur concrete, modified sulfur/sulphur concrete, durability of sulfur/sulphur concrete, etc. The basic references include the complete research timetable, in order to provide the appropriate references and milestones of the research. Fig. 9 shows the main keywords

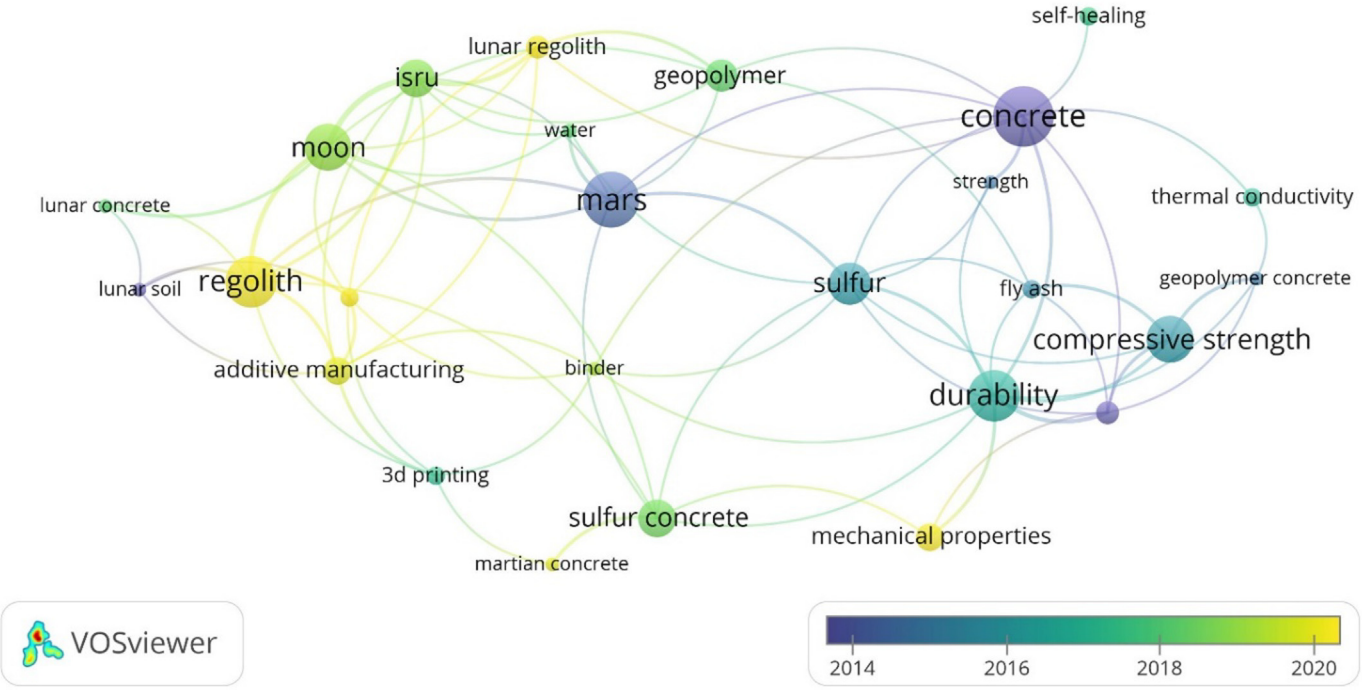

FIG. 9

Visual representation of the reference keywords concerning the sulfur concrete.

selected in the references by the VOS visualization tool. The occurrence is set to at least 3 times to filter out disturbances.

From a temporal point of view, sulfur concrete, its durability, and its mechanical performance remain important subjects for researchers. From around 2018, sulfur concrete became popular again, in connection with the Moon and Mars. Towards 2020, cutting-edge information on the regolith of the Moon and Mars was provided, and research linked to residence on-site and related construction materials was proposed, including the methodology of additive manufacturing.

From a connection point of view, sulfur concrete is still related to Martian/Lunar concrete. As mentioned in Chapter 3, sulfur is one of the components of both celestial bodies. It is promising to be used as a building material for buildings on celestial objects. Compared to compressive strength, the durability of sulfur concrete is more noteworthy, which makes sense since the extraterrestrial environment may be much harsher than that of Earth. Interestingly, self-healing, one of the most popular topics in building materials, is not closely related to sulfur concrete or extraterrestrial application, which reinforces the importance of the discussion in Chapter 4 and would be an additional point of attention.

Data availability

The data used in this review document are based on the respective literature works. All data is available.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Qinjian Wang and Didier Snoeck reports financial support was provided by Université Libre de Bruxelles - ULB.

Acknowledgments

The authors would like to thank Université Libre de Bruxelles (ULB) for funding the ARC Project « Actions de Recherche Concertées » Consolidation 2022-2025 “HeSCon: Sulphur Concrete, to enhance the structural integrity and durability of the material”.

References

[1] F. Brown, Space Policy 27 (2011) 73.
[2] Z. Hu et al., Constr. Build. Mater. 343 (2022) 128117.
[3] A Brief History of Space Exploration | The Aerospace Corporation. Aerospace Corporation. https://aerospace.org/article/brief-history-space-exploration.
[4] T.A. Press, Key dates in history of space exploration. https://phys.org/news/2012-08-key-dates-history-space-exploration.html.
[5] S. Potter, NASA Administrator Statement on President’s FY 2023 Budget Request. NASA. http://www.nasa.gov/press-release/nasa-administrator-statement-on-presidents-fy-2023-budget-request (2022).
[6] Carpenter, J. ESA Space Resources Strategy.
[7] China’s Space Program: A 2021. Perspective. http://www.cnsa.gov.cn/english/n6465645/n6465648/c6813088/content.html.
[8] B. Dunbar, FY 2020 Budget Request. NASA. http://www.nasa.gov/content/fy-2020-budget-request (2020).
[9] Mars One’s Journey. https://www.mars-one.com/#.
[10] E. Musk, New Space 6 (2018) 2–11.
[11] M.Z. Naser, Prog. Mater. Sci. 105 (2019).
[12] K.W. Farries et al., Constr. Build. Mater. 296 (2021) 123627.
[13] H.A. Omar, Production of Lunar Concrete Using Molten Sulfur. https://ntrs.nasa.gov/citations/19980001900 (1993).
[14] Y.C. Toklu, Civil Engineering in the Design and Construction of a Lunar Base. 822–834 (2012).
[15] Y.C. Toklu, P. Akpinar, Adv. Space Res. 70 (2022) 762–779.
[16] I.I. Cherkasov, V.V. Shvarev, Soil Mech. Found. Eng. 10 (1973) 252–256.
[17] I.I. Cherkasov, V.V. Shvarev, Soil Mech. Found. Eng. 14 (1977) 320–326.
[18] E.N. Slyuta, Sol. Syst. Res. 48 (2014) 330–353.
[19] J. Liu et al., Int. J. Transp. Sci. Technol. 11 (2022) 1–16.
[20] W.N. Agosto, A.M.M. Gadalla, Lunar cement formulations for space systems shielding and construction. in 179–183 (1985).
[21] Concrete — A Practical Construction Material for Mars | Engineering, Construction, and Operations in Space V. https://ascelibrary.org/doi/10.1061/40177%28207%2978.
[22] W. Khaliq, H.A. Khan, Constr. Build. Mater. 94 (2015) 475–487.
[23] M.P. Adams, J.H. Ideker, Cem. Concr. Res. 100 (2017) 284–296.
[24] Q. Ye, C.-C. Ma, Y. Li, Jianzhu Cailiao XuebaoJournal Build. Mater. 13 (2010) 654–658.
[25] J.J. O’Gallagher, Using Ultra High Solar Flux in the Lunar Environment: Production of Cement and Other Applications (2012) 571–577.
[26] J. Wei, Y. Chen, Y. Li, J. Wuhan Univ. Technol. Mater. Sci. Ed. 21 (2006) 88–91.
[27] Z. Li et al., Materials 12 (2018) 80.
[28] H.M. Tran, A. Scott, Constr. Build. Mater. 131 (2017) 526–535.
[29] Z. Li et al., J. Am. Ceram. Soc. 102 (2019) 3269–3285.
[30] H. Tran, A. Scott, R. Dhakal, Mag. Concr. Res. 72 (2020) 693–702.
[31] S.A. Walling et al., Dalton Trans. 44 (2015) 8126–8137.
[32] F. Jin, A. Al-Tabbaa, Cem. Concr. Compos. 52 (2014) 27–33.
[33] S. W. Koh, et al., Experimental study of waterless concrete for lunar construction, in: Earth and Space 2010: Engineering, Science, Construction, and Operations in Challenging Environments, 1098–1102 (2010).
[34] T.S. Lee, J. Lee, K.Y. Ann, Acta Astronaut. 114 (2015) 60–64.
[35] R. Bedi, R. Chandra, S.P. Singh, J. Compos. 2013 (2013) e948745.
[36] B. Singh et al., Constr. Build. Mater. 85 (2015) 78–90.
[37] E.I. Diaz-Loya, E.N. Allouche, S. Vaidya, ACI Mater. J. 108 (2011) 300–306.
[38] G. Xiong et al., Constr. Build. Mater. 325 (2022) 126679.
[39] K. Wang et al., Sci. Rep. 6 (2016) 29659.
[40] C. Montes et al., Space Res. 56 (2015) 1212–1221.
[41] S. Wilhelm, M. Curbach, Struct. Concr. 15 (2014) 419–428.
[42] M.M. Vlahovic et al., Constr. Build. Mater. 25 (2011) 3926–3934.
[43] A. Khitab et al., Astron. Rep. 60 (2016) 306–312.
[44] R. S. Leonard, & S. W. Johnson, Sulfur-Based Construction Materials for Lunar Construction. in 1295–1307 (ASCE, 1988).
[45] A.G. Khademi, H.I.K. Sar, Curr World Env. 10 (2015) 63–68.
[46] R.N. Grugel, H. Toutanji, Adv. Space Res. 41 (2008) 103–112.
[47] R.N. Grugel, Adv. Space Res. 50 (2012) 1294–1299.
[48] L. Wan, R. Wendner, G. Cusatis, Constr. Build. Mater. 120 (2016) 222–231.
[49] A. Barkatt, M. Okutsu, J. Chem. Res. 46 (2022) 17475198221080729.
[50] Y. Abdel-Jawad, M. Al-Qudah, Cem. Concr. Res. 24 (1994) 165–175.
[51] M. Dugarte, G. Martinez-Arguelles, J. Torres, Sustainability 11 (2018) 70.
[52] R. Fediuk et al., Materials 13 (2020) 1–23.
[53] E. K. Gibson, Jr, & G. W. Moore, Carbon and sulfur distributions and abundances in lunar fines, in: Proceedings of the Lunar Science Conference, vol. 4, 1973, p. 1577.
[54] D. Vaniman, D. Pettit, G. Heiken, Uses of lunar sulfur, in: NASA. Johnson Space Center, The Second Conference on Lunar Bases and Space Activities of the 21st Century, Volume 2, 1992.
[55] D.T. Vaniman, G.H. Heiken, Eng. Constr. Oper. Space II (1) (1990) 107–116.
[56] J.J. O’Gallagher, T.D. Lin, Spec. Publ. 125 (1991) 245–254.
[57] H. Toutanji, B. Glenn-Loper, B. Schrayshuen, Strength and durability performance of waterless lunar concrete, 43rd AIAA Aerospace Sciences Meeting and Exhibit 1436, 2005.
[58] L. Taylor et al., Adv. Space Res. 61 (2018) 1490–1500.
[59] A. Alexiadis, F. Alberini, M.E. Meyer, Adv. Space Res. 59 (2017) 490–495.
[60] S. Seitl, et al. Mechanical fracture properties of concrete with lunar aggregate simulant, in: MATEC Web of Conferences, vol. 323, 01014 (EDP Sciences, 2020).
[61] G. Cesaretti et al., Acta Astronaut. 93 (2014) 430–450.
[62] C. Buchner et al., Acta Astronaut. 143 (2018) 272–284.
[63] Guide for Mixing and Placing Sulfur Concrete in Construction. ACI Mater. J. 85 (1988).
[64] W.L. Sheppard Jr, Sulphur Inst. J. 11 (1975).
[65] W.H. Kobbe, Ind. Eng. Chem. 16 (1924) 1026–1028.
[66] W.S. Rybczynski, Concrete and Very Low Cost Housing. (1974).
[67] W.C. McBee, T.A. Sullivan, H.L. Fike, Sulfur construction materials, US Department of the Interior, Bureau of Mines, 1985.
[68] V. Gracia, E. Vàzquez, S. Carmona, Utilization of by-produced sulfur for the manufacture of unmodified sulfur concrete, in: International RILEM

Conference on the Use of Recycled Materials in Buildings and Structures vol. 1 1–2 (2004).
[69] R.F. Bacon, H.S. Davis, Recent advances in the American sulphur industry. (1921).
[70] Annual report of the Secretary of the Interior under the Mining and Minerals Policy Act of 1970. Annu. Rep. Secr. Inter. Min. Miner. Policy Act 1970 (1972).
[71] R.E. Loov, A.H. Vroom, M.A. Ward, PCI J. 19 (1974) 86–95.
[72] J.J. Beaudoin, P.J. Sereda, Freeze-thaw Durability of Sulfur Concrete, National Research Council Canada, Division of Building Research, 1974.
[73] V. M. Malhotra, Effect of Specimen Size on compressive strength of sulfur concrete. Division Report No. IR 74-25. Energy Mines Resour. Can. Ottwa (1974).
[74] V.M. Malhotra, Mechanical properties and freeze-thaw resistance of sulphur concrete, Department of Mines and Technical Surveys, 1973.
[75] W.C. McBee, T.A. Sullivan, B.W. Jong, NASA STIRecon Tech. Rep. N 82 (1981) 11333.
[76] W.C. McBee, T.A. Sullivan, B.W. Jong, Corrosion-Resistant Sulfur Concretes, vol. 8758, US Department of the Interior, Bureau of Mines, 1983.
[77] W.C. McBee, T.A. Sullivan, B.W. Jong, Industrial evaluation of sulfur concrete in corrosive environments, US Department of the Interior, Bureau of Mines, 1983.
[78] R.H. Funke Jr, W.C. McBee, An industrial application of sulfur concrete, ACS Publications, 1982.
[79] W.C. McBee, T.A. Sullivan, Development of Specialized Sulfur Concretes, vol. 8346, Department of the Interior, Bureau of Mines, 1979.
[80] F. Kesicki, Energy Policy 38 (2010) 1596–1606.
[81] P. R. et al. A life cycle analysis of its health and environmental impacts. Cent. Health Glob. Environ. Harv. Medial Sch. EUA (2002).
[82] N. Sakib et al., Int. J. Pavement Eng. 22 (2021) 392–403.
[83] J. Quik, E. Dekker, M. Montforts, Safety and sustainability analysis of railway sleeper alternatives . Application of a novel method for material loops. https://rivm.openrepository.com/handle/10029/624641 (2020) doi:10.21945/RIVM-2020-0181.
[84] STARcrete™ - stable, acid resistant, sulfur-based concrete, extreme corrosion resistance, high physical strength, low water permeability. http://www.starcrete.com/.
[85] J.-G. Wagenfeld et al., Waste Manag. 95 (2019) 78–89.
[86] First recyclable sulfur concrete sleepers placed in Belgium. RailTech.com https://www.railtech.com/infrastructure/2021/03/08/first-recyclable-sulfur-concrete-sleepers-placed-in-belgium/ (2021).
[87] Cement - prices by country, around the world, September 2023 | GlobalProductPrices.com. https://www.globalproductprices.com/rankings/cement_prices/.
[88] Contact us via LiveChat! https://secure.livechatinc.com/.
[89] D. Husam, A. Omar, Final Research Report for JoVe NASA Grant NAG8 - 278.
[90] P. Raju, Int. J. Eng. Sci. Adv. Technol. 4 (2014) 501–510.
[91] J.T. Burwell, Z. Für Krist.-Cryst. Mater. 97 (1937) 123–124.
[92] M. Thackray, J. Chem. Eng. Data 15 (1970) 495–497.
[93] A.D. Smith, R.C. Smith, A.G. Tennyson, Sustain. Chem. Pharm. 16 (2020) 100249.
[94] H. Benaroya, Lunar settlements, CRC Press, 2010.
[95] M.A. Gulzar et al., J. Build. Eng. 38 (2021) 102175.
[96] I. Casanova, V. Aulesa, Space 2000 (2000) 638–644.
[97] P. Szajerski et al., J. Clean. Prod. 271 (2020) 122563.
[98] S. Al-Otaibi et al., J. King Saud Univ.-Eng. Sci. 31 (2019) 327–331.
[99] R. Anyszka et al., Sulfur concrete-promising material for space-structures building, in: Proceedings of the European Conference on Spacecraft Structures Materials and Environmental Testing, 2016, pp. 27–30.
[100] S.G. Bae et al., KSCE J. Civ. Environ. Eng. Res. 34 (2014) 763–771.
[101] S.H. Deshmukh, J.P. Bhusari, A.M. Zende, IOSR J. Eng. 2 (2012) 1308–1312.
[102] Q. Tang et al., Cem. Concr. Compos. 114 (2020) 103807.
[103] Z. Duan et al., Constr. Build. Mater. 254 (2020) 119323.
[104] J.D. Birchall, A.J. Howard, K. Kendall, Nature 289 (1981) 388–390.
[105] Y. Shen et al., Constr. Build. Mater. 229 (2019) 116893.
[106] M.M. Vlahović et al., Mater. Des. 34 (2012) 346–354.
[107] S. Tsivilis et al., Cem. Concr. Res. 33 (2003) 1465–1471.
[108] W.R. Wrzesinski, W.C. McBee, Permeability and Corrosion Resistance of Reinforced Sulfur Concrete, vol. 9157, US Department of the Interior, Bureau of Mines, 1988.
[109] G. R. Darnell, Sulfur polymer cement, a new stabilization agent for mixed and Low-level radioactive waste. (1991).
[110] A. Dey et al., Materials 14 (2021) 254.
[111] I. Asadi et al., J. Build. Eng. 20 (2018) 81–93.
[112] T.S. Yun et al., Energy Build. 61 (2013) 125–132.
[113] D. Kraynov, G. Medvedeva, E3S Web Conf. 274 (2021) 07001.
[114] N. Ciak, J. Harasymiuk, Tech. Sci. Warm. Mazury Olszt. 323–331 (2013).
[115] R.B. Malla, K.M. Brown, Acta Astronaut. 107 (2015) 196–207.
[116] M.R. Meier et al., Constr. Build. Mater. 93 (2015) 877–883.
[117] M. Schönlein, J. Plank, J. Am. Ceram. Soc. 101 (2018) 3575–3584.
[118] J. Moraes Neves et al., Front. Mater. 6 (2019) 83.
[119] C. Meyers, H. Toutanji, J. Aerosp. Eng. 20 (2007) 220–226.
[120] H. Toutanji, B. Schrayshuen, M. Han, New glass fiber reinforced concrete for extraterrestrial application, in: Earth & Space 2006: Engineering, Construction, and Operations in Challenging Environment 1–8 (2006).
[121] S. Mohammed, V. Poornima, Mater. Today Proc. 5 (2018) 23888–23897.
[122] M. Książek, Proc. Inst. Civ. Eng. - Constr. Mater. 170 (2017) 281–296.
[123] A.-M.O. Mohamed, M. El-Gamal, Sulfur concrete for the construction industry: a sustainable development approach, J. Ross Publishing, 2010.
[124] M.M. El Gamal et al., J. Build. Eng. 11 (2017) 1–8.
[125] M.H. Shahsavari et al., Constr. Build. Mater. 350 (2022) 128914.
[126] S. Ghasemi et al., J. Mater. Civ. Eng. 34 (2022).
[127] M. Troemner, et al., Marscrete: A Martian Concrete for Additive Construction Applications Utilizing In Situ Resources. 807 (2021).
[128] B. Chashm Khavari et al., Constr. Build. Mater. 384 (2023) 131434.
[129] A.H. Vroom, Sulphur Utilization: A Challenge and an Opportunity, National Research Council of Canada, 1971.
[130] J.M. Dale, A.C. Ludwig, Civ. Eng. (1967).
[131] R. Gregor, A. Hackl, A new approach to sulfur concrete, ACS Publications, 1978.
[132] M. Sabour, G. Dezvareh, R. Bazzazzadeh, Constr. Build. Mater. 199 (2019) 40–49.
[133] D. Yeoh, K.H. Boon, N. Jamaluddin, J. Teknol. 77 (2015).
[134] F.A. López Gómez, et al., The application of sulfur concrete to the stabilization of Hg-contaminated soil. (2009).
[135] H.A. Toutanji, S. Evans, R.N. Grugel, Constr. Build. Mater. 29 (2012) 444–448.
[136] B. Khoshnevis, et al. Lunar Contour Crafting - A Novel Technique for ISRU-Based Habitat Development, in: 43rd AIAA Aerospace Sciences Meeting and Exhibit, American Institute of Aeronautics and Astronautics.
[137] N.J. Bennett, D. Ellender, A.G. Dempster, Planet. Space Sci. 182 (2020) 104842.
[138] G.H. Heiken, D.T. Vaniman, B.M. French, Lunar Sourcebook, a user's guide to the Moon. (1991).
[139] C. Nunn et al., Space Sci. Rev. 216 (2020) 89.
[140] H.J. Melosh, Planetary Surface Processes, vol. 13, Cambridge University Press, 2011.
[141] P. Lucey et al., Rev. Mineral. Geochem. 60 (2006) 83–219.
[142] I.A. Crawford, Prog. Phys. Geogr. Earth Environ. 39 (2015) 137–167.
[143] B.A.C.S. Jayathilake, I.M.S.K. Ilankoon, M.N.P. Dushyantha, Acta Astronaut. 196 (2022) 107–122.
[144] W.D. Carrier III, J. Geotech. Geoenvironmental Eng. 129 (2003) 956–959.
[145] P.K. Zarzycki, J. Katzer, J. Aerosp. Eng. 32 (2019) 06019005.
[146] M.R. Ponnada, P. Singuru, Int J Eng Sci Adv Technol 4 (2014) 501–510.
[147] S. Lee, A. van Riessen, Materials 15 (2022) 4516.
[148] T. Chen et al., Adv. Space Res. 61 (2018) 830–836.
[149] H. Su et al., Sci. Rep. 9 (2019) 2238.
[150] K. Oh et al., CEAS Space J. 13 (2021) 183–187.
[151] J.N. Mills, M. Katzarova, N.J. Wagner, Adv. Space Res. 69 (2022) 761–777.
[152] G. Scheerbaum, Space 2000 (2000) 934–940.
[153] J. Gómez-Elvira et al., Space Sci. Rev. 170 (2012) 583–640.
[154] R. Sullivan et al., Nature 436 (2005) 58–61.
[155] V.R. Baker, Nature 412 (2001) 228–236.
[156] J.I. Lunine et al., Icarus 165 (2003) 1–8.
[157] R.M. Haberle et al., J. Geophys. Res. Planets 106 (2001) 23317–23326.
[158] M. Nazari-Sharabian et al., Galaxies 8 (2020) 40.
[159] G.A. Morgan et al., Nat. Astron. 5 (2021) 230–236.
[160] Chinese rover makes surprise discovery about liquid water on Mars. Big Think https://www.freethink.com/space/liquid-water-on-mars (2022).
[161] P.L. King, S.M. McLennan, Elements 6 (2010) 107–112.
[162] F. Gaillard et al., Space Sci. Rev. 174 (2013) 251–300.
[163] K. Lodders, B. Fegley, Icarus 126 (1997) 373–394.
[164] Z. Wang, H. Becker, Earth Planet. Sci. Lett. 463 (2017) 56–68.
[165] A. Khan, J.A.D Connolly, J. Geophys. Res. Planets 113 (2008).
[166] C. Li et al., Space Sci. Rev. 217 (2021) 57.
[167] O. Sharaf, Nat. Astron. 4 (2020) 722.
[168] T. Polsgrove, et al. Transit Habitat Design for Mars Exploration, in: 2018 AIAA SPACE and Astronautics Forum and Exposition 5143 (2018).

[169] D. S. McKay, C. C. Allen, Concrete—a practical construction material for Mars, in Engineering, Construction, and Operations in Space V 566–570 (1996).
[170] H. Li et al., Powder Technol. 390 (2021) 219–229.
[171] Soil on Mars. Let's Talk Science https://letstalkscience.ca/educational-resources/backgrounders/soil-on-mars (2021).
[172] A. Scott, C. Oze, M.W. Hughes, J. Aerosp. Eng. 33 (2020) 04020019.
[173] J. Karni, E. Karni, Mater. Struct. 28 (1995) 92–100.
[174] F.A. Blakey, Archit. Sci. Rev. 4 (1961) 6–16.
[175] Y. Reches, Cem. Concr. Compos. 104 (2019) 103349.
[176] P.B. Niles et al., Space Sci. Rev. 174 (2013) 301–328.
[177] B. Kading, J. Straub, Acta Astronaut. 107 (2015) 317–326.
[178] R. Kelso, AGU Fall Meeting Abstracts 2013 (2013) V53C-2808.
[179] Elemental Composition of the Martian Crust | Science. https://www.science.org/doi/10.1126/science.1165871.
[180] Y. Wang et al., Appl. Clay Sci. 229 (2022) 106673.
[181] Q. Sun et al., Appl. Clay Sci. 199 (2020) 105870.
[182] M. Isachenkov et al., Acta Astronaut. 180 (2021) 650–678.
[183] J. Xu, et al., 3D printing of hypothetical brick by selective laser sintering using lunar regolith simulant and ilmenite powders, in: 9th Int Symp Adv Opt Manuf Test Technol Subdiffraction-Ltd. Plasmonic Lithogr Innov Manuf Technol 10842 (2019) 38–48.
[184] M. Anand et al., Planet. Space Sci. 74 (2012) 42–48.
[185] H.M. Sargeant et al., Planet. Space Sci. 205 (2021) 105287.
[186] J.A. Hoffman et al., Sci. Adv. 8 (2022) eabp8636.
[187] G.H. Just et al., Planet. Space Sci. 180 (2020) 104746.
[188] D. L. Linne, et al. Overview of NASA Technology Development for In-Situ Resource Utilization (ISRU). in (2017).
[189] W.L. Johnson et al., Cryogenics 90 (2018) 60–69.
[190] S.S. Alam et al., iScience 25 (2022) 104323.
[191] G. B. Sanders, et al., Mars ISRU for Production of Mission Critical Consumables - Options, Recent Studies, and Current State of the Art, in: AIAA SPACE 2015 Conference and Exposition (American Institute of Aeronautics and Astronautics). doi:10.2514/6.2015-4458.
[192] B. Khoshnevis, A. Carlson, M. Thangavelu, ISRU-Based Robotic Construction Technologies For Lunar And Martian Infrastructures. (2017).
[193] G. Sanders, J. Kleinhenz, In Situ Resource Utilization (ISRU) Envisioned Future Priorities.
[194] R.M. Fruland, J.N. Reimold, Introduction to the Core Samples from the Apollo 16 Landing Site, National Aeronautics and Space Administration, Lyndon B. Johnson Space Center, 1981.
[195] R.V. Morris, Handbook of lunar soils, Lyndon B. Johnson Space Center, 1983.
[196] mars.nasa.gov. NASA's Perseverance Rover Gets the Dirt on Mars. NASA Mars Exploration https://mars.nasa.gov/news/9311/nasas-perseverance-rover-gets-the-dirt-on-mars.
[197] D. Karl, K.M. Cannon, A. Gurlo, Open Ceram. 9 (2022) 100216.
[198] T.D. Lin, L. Tseng, S. Chou, Lunar concrete made with the dry-mix/steam-injection method, in: Engineering, Construction, and Operations in Space V 592–599 (1996).
[199] P.-M. Kost et al., Acta Astronaut. 187 (2021) 429–437.
[200] A. Zocca et al., Ceram. Int. 46 (2020) 14097–14104.
[201] K. Bochenek, M. Basista, Prog. Aerosp. Sci. 79 (2015) 136–146.
[202] G.B. Sanders, W.E. Larson, J. Aerosp. Eng. 26 (2013) 5–17.
[203] published, J. G.-D. Chandrayaan-3's measurements of sulfur open the doors for lunar science and exploration. *Space.com* https://www.space.com/chandrayaan-3-sulfur-measurements-lunar-science-expert-voices (2023).
[204] X. Yuan, et al. Performance of Sulfur Concrete in Planetary Applications of Contour Crafting. (2016).
[205] A.-M.-O. Mohamed, M.E. Gamal, Cem. Concr. Compos. 31 (2009) 186–194.
[206] Z. Yang et al., Chem. Eur. J. 25 (2019) 5643–5647.
[207] S.-L. Lin, J.S. Lai, E.S.K. Chian, Waste Manag. 15 (1995) 441–447.
[208] B.R. Currell, Symposium on New Users for Sulfur and Pyrites, Madrid (1976) 105e113.
[209] B.K. Bordoloi, E.M. Pearce, Plastic sulfur stabilization by copolymerization of sulfur with dicyclopentadiene, ACS Publications, 1978.
[210] J.J. Beaudoin, R.F. Feldman, Int. J. Cem. Compos. Lightweight Concr. 6 (1984) 13–18.
[211] E.L.C. Nnabuife, Indian J. Technol. 25 (1987) 363–367.
[212] A.-M.-O. Mohamed et al., Environ. Geotech. 2 (2015) 95–103.
[213] A.-M.-O. Mohamed, M. El Gamal, Environ. Geol. 53 (2007) 159–175.
[214] A.M.O. Mohamed, M.M. El Gamal, A.K. El Saiy, Thermo-mechanical behavior of newly developed sulfur polymer concrete, in: 27–38 (2006).
[215] D. Snoeck et al., Mater. Struct. 49 (2016) 225–239.
[216] R. Baetens, B.P. Jelle, A. Gustavsen, Energy Build. 42 (2010) 1361–1368.
[217] K. Faraj et al., Renew. Sustain. Energy Rev. 119 (2020) 109579.
[218] D.P. Bentz, R. Turpin, Cem. Concr. Compos. 29 (2007) 527–532.
[219] M. Delgado et al., Renew. Sustain. Energy Rev. 16 (2012) 253–273.
[220] K. Du et al., Appl. Energy 220 (2018) 242–273.
[221] V.D. Cao et al., Energy Convers. Manag. 133 (2017) 56–66.
[222] X. Bao et al., Materials 10 (2017) 462.
[223] X. Bao et al., Energy Build. 194 (2019) 33–45.
[224] B. Koohestani et al., Cem. Concr. Compos. 72 (2016) 180–189.
[225] B.A. Newcomb, Compos. Part Appl. Sci. Manuf. 91 (2016) 262–282.
[226] X. Liu et al., Renew. Energy 218 (2023) 119262.
[227] Z.A. Al-Absi, M.I.M. Hafizal, M. Ismail, Energy Build. 288 (2023) 113003.
[228] M. Solaimanian, T.W. Kennedy, Transp. Res. Rec. (1993).
[229] P. Chestovich et al., J. Burn Care Res. 44 (2022).
[230] X. Gong et al., Case Stud. Constr. Mater. 18 (2023) e01852.
[231] N. De Belie et al., Adv. Mater. Interfaces 5 (2018) 1800074.
[232] N.Z. Muhammad et al., Constr. Build. Mater. 112 (2016) 1123–1132.
[233] H. Huang, G. Ye, D. Damidot, Cem. Concr. Res. 60 (2014) 68–82.
[234] D. Snoeck, L. Pel, N. De Belie, Sci. Rep. 7 (2017) 1–14.
[235] A.M. Abd_Elmoaty, Alex. Eng. J. 50 (2011) 171–178.
[236] L. Ferrara et al., Constr. Build. Mater. 167 (2018) 115–142.
[237] H.M. Jonkers et al., Ecol. Eng. 36 (2010) 230–235.
[238] P. Minnebo et al., Materials 10 (2017) 49.
[239] J.Y. Wang et al., Constr. Build. Mater. 68 (2014) 110–119.
[240] S. Gwon, E. Ahn, M. Shin, Compos. Part B Eng. 162 (2019) 469–483.
[241] D. Snoeck, Infrastructures 7 (2022) 129.
[242] D. Snoeck, L. Pel, N. De Belie, Sci. Rep. 10 (2020) 642.
[243] D. Snoeck, N. De Belie, J. Mater. Civ. Eng. 28 (2016) 04015086.
[244] D. Snoeck et al., J. Intell. Mater. Syst. Struct. 25 (2014) 13–24.
[245] D. Snoeck, N. De Belie, Constr. Build. Mater. 95 (2015) 774–787.
[246] P. Bazin, J. Saunier, Deformability, fatigue and healing properties of asphalt mixes. in Intl Conf Struct Design Asphalt Pvmts (1967).
[247] S. Xu et al., Adv. Mater. Interfaces 5 (2018) 1800536.
[248] P. Ayar, F. Moreno-Navarro, M.C. Rubio-Gámez, J. Clean. Prod. 113 (2016) 28–40.
[249] Q. Lv et al., Mater. Des. 117 (2017) 7–17.
[250] à. García, Fuel 93 (2012) 264–272.
[251] A. Menozzi et al., Constr. Build. Mater. 74 (2015) 162–168.
[252] à. García et al., Constr. Build. Mater. 23 (2009) 3175–3181.
[253] Y. Sun et al., Constr. Build. Mater. 150 (2017) 673–680.
[254] M.A. Franesqui, J. Yepes, C. García-González, Constr. Build. Mater. 149 (2017) 612–620.
[255] à. García et al., Constr. Build. Mater. 31 (2012) 38–46.
[256] Q. Liu, E. Schlangen, M. van de Ven, J. Mater. Civ. Eng. 25 (2013) 880–885.
[257] J. Gallego et al., Constr. Build. Mater. 42 (2013) 1–4.
[258] Q. Liu et al., Constr. Build. Mater. 24 (2010) 1207–1213.
[259] A. Garcia, J. Jelfs, C.J. Austin, Constr. Build. Mater. 101 (2015) 309–316.
[260] J.-F. Su et al., Constr. Build. Mater. 74 (2015) 83–92.
[261] J.-F. Su et al., Mater. Struct. 48 (2015) 4067–4076.
[262] A. Garcia, C.J. Austin, J. Jelfs, J. Clean. Prod. 118 (2016) 124–132.
[263] X. Wang et al., J. Smart Cities 2 (2019) 1–39.